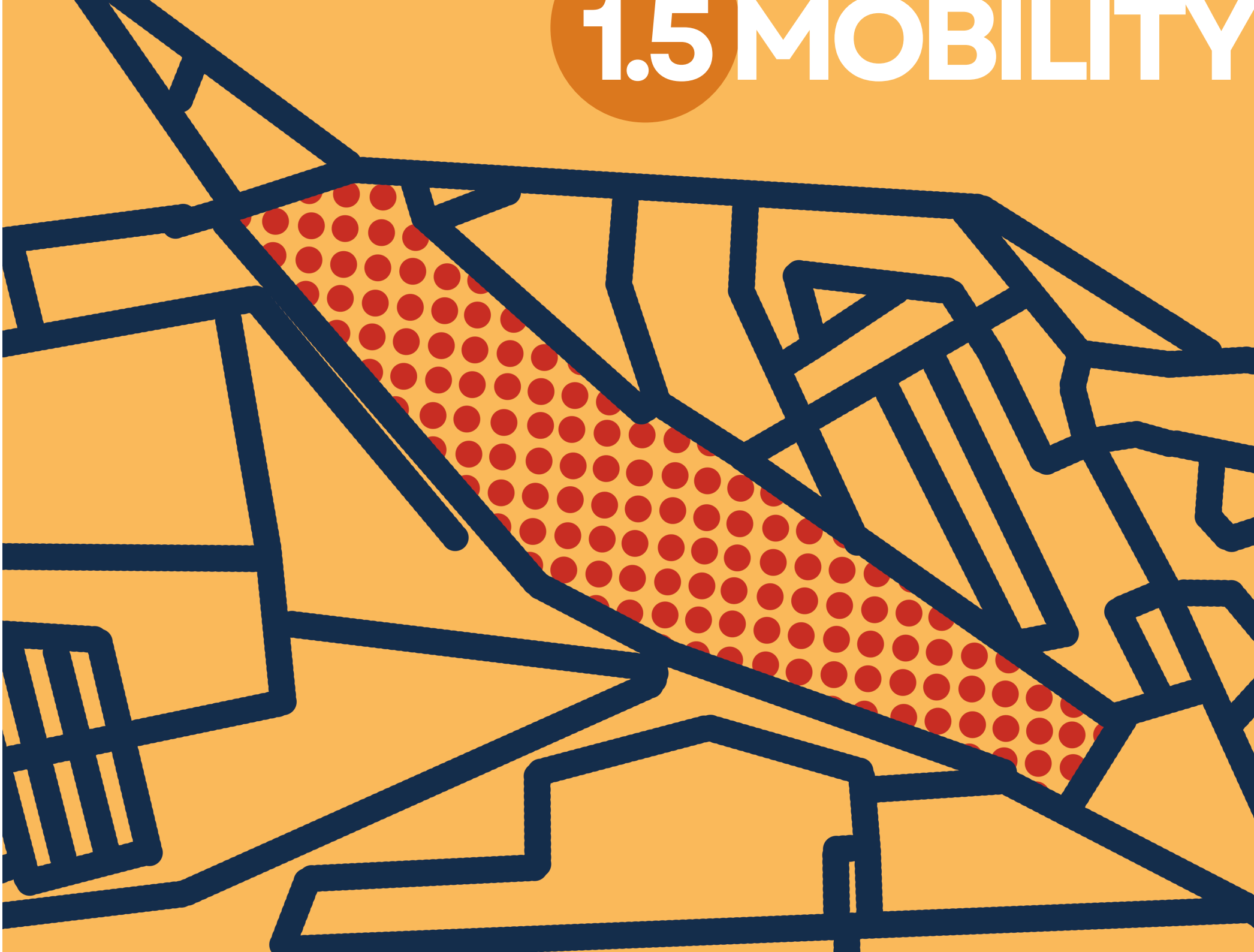
Metropolitan Solutions

MADE 2025

1.5 MOBILITY



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ABOUT THE EXPERT TEAM

All members of this team have different backgrounds, making this team diverse in its skills and knowledge. During this project, we used this to our advantage. We have discussed a lot about our vision for the project and aligned all our ideas to create a report that truly includes everyone's strengths and skills. We see ourselves as five different expert lenses, creating one harmonized report.

All maps, figures, and tables are made by the authors, unless otherwise specified.

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MOBILITY

As cities aim to reduce car dependency, car-free neighbourhoods are gaining attention as a pathway to more sustainable and livable urban environments. This report investigates the potential of transforming Zaanstraat Emplacement into a car-free neighbourhood while ensuring it remains accessible, inclusive, and well-connected.

Through a theoretical framework supported by spatial analysis and expert insights, the report identifies critical mobility challenges and opportunities. While car-free design can risk reducing accessibility, connectivity, and inclusiveness if poorly implemented, these barriers can be overcome through high-quality public transport, integrated cycling networks, and inclusive street design. The report ends with a reflection on the Zaanstraat Emplacement and concludes with key conditions under which a car-free development of the area can be realistically achieved.

1 INTRODUCTION

Car usage is associated with numerous negative externalities, such as greenhouse gas emissions, congestion, accidents, and noise pollution (Hensher, 2008; Magdin et al., 2019). In response, urban planners and policymakers have increasingly aimed to reduce the reliance on private cars and stimulate more sustainable forms of mobility. One commonly used strategy in the Netherlands is a low-car approach, which seeks to reduce car usage and ownership while still permitting low levels of car access and use (Jorritsma, 2023). However, because harmful externalities are directly tied to the presence and use of motor vehicles, a car-free neighbourhood design could offer a more effective solution. By eliminating private cars from the living environment, such designs can more comprehensively address the spatial, environmental, and social challenges associated with car ownership and usage compared to conventional low-car strategies (Bertolini, 2017).

Despite these benefits, going fully car-free comes with numerous challenges in its implementation. Nieuwenhuijsen et al. (2019) identified nine prerequisites required in the transition to car-free environments, such as: 'Stakeholder support and participation', 'alternative convenient and quality transport means', and 'a detailed plan aligned with other high-level objectives'.

A failure to meet these preconditions could lead to unsatisfactory results: firstly, without strong local support, car-free developments may lack inclusivity as they fail to take the mobility needs of diverse user groups into account, such as those with disabilities, the elderly or low-income residents (Carvalho et al., 2024). Secondly, if no alternative transport methods are available to fill the gap that cars leave, accessibility is negatively impacted, resulting in destinations being harder to reach (Basu & Ferreira, 2021; Nielsen et al., 2013; Sun et al., 2017). Lastly, without a detailed plan aligned with higher-level objectives that transcends the neighbourhood level scale, a car-free approach is at risk of becoming poorly integrated with the surrounding networks, jeopardising greater metropolitan connectivity (Nieuwenhuijsen et al., 2019).

Therefore, when considering a car-free approach, it is essential to ensure that the principles of inclusivity, accessibility, and well-connectedness are not compromised, as they are critical to creating a functional, equitable, and integrated living environment.

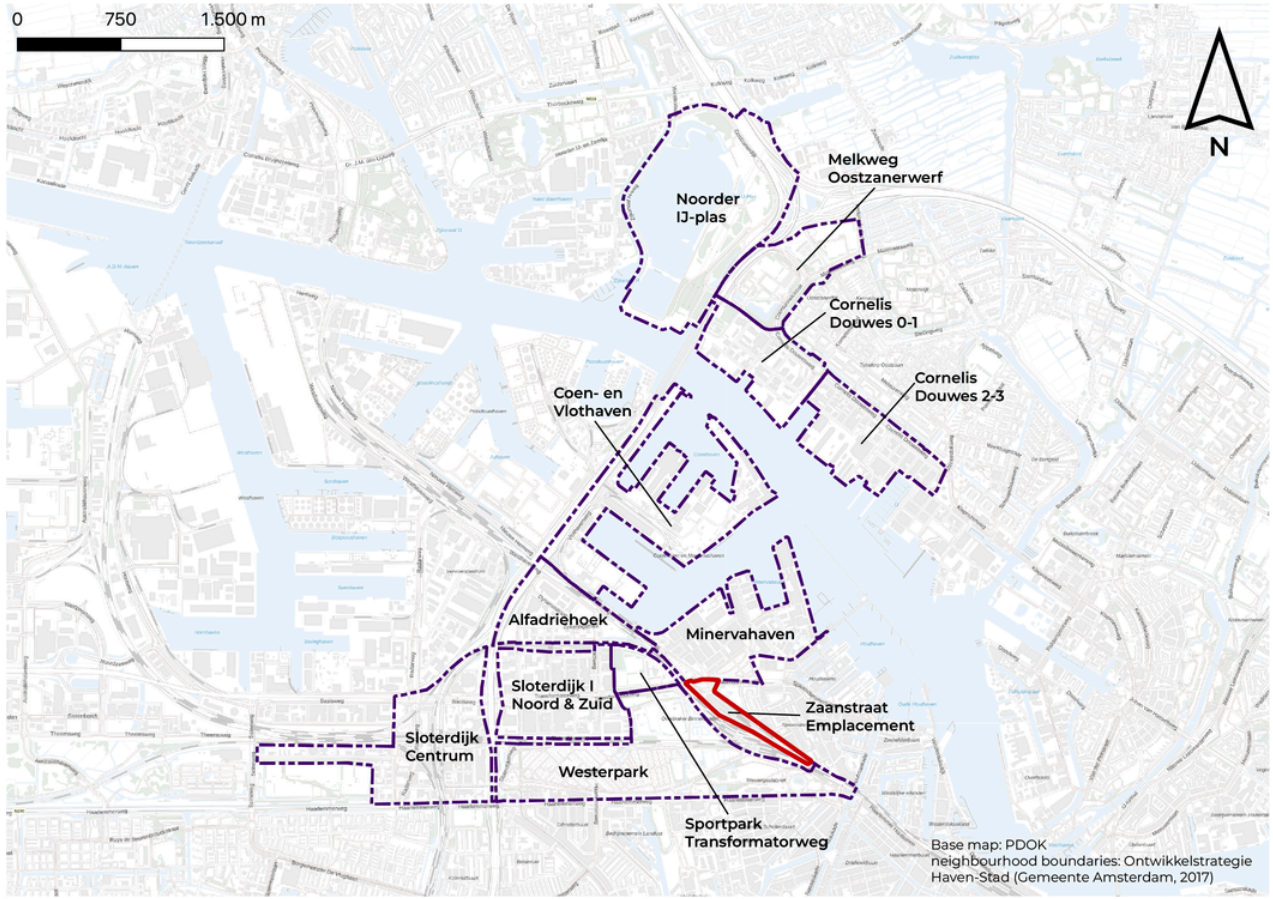
1.1 STUDY AREA CONTEXT

In this research, Zaanstraat Emplacement functions as the study area for exploring the potential of car-free neighbourhood design. This neighbourhood is situated in Haven-Stad, which is located in the northwestern part of Amsterdam and encompasses a vast area of former docks, port areas, and industrial estates. Its location within the A10 ring motorway, proximity to the historic town centre, and its connection to major transport corridors make it a strategic site for urban redevelopment.

The municipality of Amsterdam plans to gradually transform the area into a high-density, mixed-use urban neighbourhood within the coming 35 years.

Based on the development plans, Haven-Stad is expected to house between 40.000 to 70.000 new dwellings and 45.000 to 58.000 jobs, which makes it the largest inner-city transformation project in the Netherlands (Gemeente Amsterdam, 2021a; van Dort, 2022).

This projected growth will lead to a significant increase in the number of residents, workers, and commuters travelling within, to, and from Haven-Stad, which will put further strain on the existing transport infrastructure.



Map 1: Haven-Stad area with Zaanstraat Emplacement highlighted in red, adapted from Ontwikkelvisie Haven-Stad 2017.

1.2 RELEVANCE

The Municipality of Amsterdam's envisioned mobility strategy for Haven-Stad aims to limit parking availability and reduce car use among residents. Melia (2014) identifies these two policy choices as features of low-car development, whereas he defines car-free development as a traffic-free environment, with no to limited parking separated from residences, and designed to enable residents to live without owning a car. Other cities have adapted this more ambitious approach, such as the Merwedekanaalzone, situated in Utrecht and currently being redeveloped from an office and industrial area into a dense waterfront neighbourhood. Private cars will not be allowed within the neighbourhood itself; instead, residents and visitors are required to park in underground parking facilities located at the outer edges of the area (Gemeente Utrecht, 2021). It is set to become the biggest neighbourhood in the Netherlands with a car-free public space, housing approximately 6000 dwellings. The Merwedekanaalzone case demonstrates that large-scale car-free development has been applied in the Dutch context.

Therefore, the smaller area, Zaanstraat Emplacement, would offer a valuable opportunity to apply similar principles in Amsterdam. By exploring the implications of car-free development in Zaanstraat Emplacement, this study offers an insight into how car-free strategies might influence the accessibility, well-connectedness, and inclusivity of the area.

There is a limited amount of academic literature regarding the evaluation of the consequences of car-free neighbourhoods. While multiple studies have explored the associated benefits of the principle (Glazener et al., 2022; Khreis & Nieuwenhuijsen, 2016) or analysed global cases to identify recommendations for successful implementation (Doheim et al., 2020), there remains limited empirical research on its implementation, results, and social acceptance on a neighbourhood level in the Dutch context. Given its early position in the Haven-Stad development schedule and its favourable urban characteristics, Zaanstraat Emplacement could offer a valuable opportunity to address this gap through empirical research.

2 RESEARCH OBJECTIVES

This research aims to gain insights into the extent to which the Zaanstraat Emplacement neighbourhood could be transformed according to the principles of car-free development. This site currently accommodates a railyard and railway workshop and will be one of the first areas within Haven-Stad to be transformed. Its strategic location near two planned metro stations makes it an ideal site for this pilot experiment (Gemeente Amsterdam, 2021b). The research aims to evaluate to what extent Zaanstraat Emplacement can serve as a pilot case for testing car-free design principles, offering valuable insights for future applications in the greater Haven-Stad developments.

The vision of this report is to reimagine the Zaanstraat Emplacement as a fully car-free neighbourhood while addressing the associated risks of losing accessibility, inclusivity and well-connectedness. For this, it employs the academic definition of the term 'car-free' as defined by Melia (2014): Residential or mixed-use neighbourhoods that provide a living environment free of traffic, limited to no parking that is separated from residences, and are designed to enable living without owning a car.

This definition will serve as the conceptual basis for this report. However, for this study, its scope is further narrowed to exclusively focus on personal mobility, i.e. the mobility of Zaanstraat Emplacement residents. The mobility needs of other actors, such as businesses, emergency services, and parcel deliveries, are excluded from the analysis.

The research question of this study is as follows:

To what extent can the Zaanstraat Emplacement be redeveloped as a car-free neighbourhood while remaining accessible, inclusive, and well-connected within the Haven-Stad area?

To comprehensively answer the main research question, it is supported by the following sub-questions:

1. **What are the (pre-)conditions and barriers to implement a car-free neighbourhood in the Zaanstraat Emplacement area?**
2. **How suitable is the planned infrastructure in and around the Zaanstraat Emplacement for supporting walking, cycling, and public transport as primary modes of mobility?**
3. **What policy and governance solutions can be implemented to ensure that residents can live/work in Zaanstraat emplacement without relying on cars?**

The remainder of this report is organised as follows. Chapter 3 provides a structured foundation of the concepts and theories within this research. In Chapter 4, the methodological methods used in this research are clarified. Chapter 5 focuses on the analyses performed. In Chapter 6, the limitations and future research opportunities are discussed. The final chapter concludes and gives recommendations to successfully implement a car-free Zaanstraat Emplacement.

3 THEORETHICAL FRAMEWORK

The following chapter provides a theoretical foundation for this research study by discussing why and how a car-free development approach can serve as an instrument to make cities more sustainable. It does so by exploring possible measures and interventions that can be implemented to create car-free living environments. Lastly, it addresses and explains the three main mobility goals of the area: accessibility, inclusivity, and well-connectedness, and investigates the relation between these concepts and a car-free neighbourhood design.

3.1 CAR-FREE NEIGHBOURHOOD

Urban transport systems have a great impact on the spatial form and liveability of cities. Of all transport modes, the car has had the greatest impact on city space, as it consumes significant amounts of land for related functions, such as roads and parking facilities. This has come at the cost of green spaces, communal areas, and housing potential (Bertolini, 2017; Verkade & Brömmelstroet, 2024). Car-free neighbourhoods reverse this logic: the removal of cars opens up space for walking, cycling, community interaction, and ecological functions (Bertolini, 2017).

Therefore, a car-free design could not only enhance the use of more sustainable forms of mobility but also reclaim and rehumanise public spaces (Gehl, 2013).

The removal of cars from neighbourhoods enables urban design to return to the human scale. Gehl (2013) argues that cities ought to be designed for people, rather than cars. Therefore, walkability, sociability, and sensory engagement should be guiding the urban design of cities (Gehl, 2013). Car-free environments align with this vision by prioritising public spaces that foster face-to-face interaction, community cohesion, and active lifestyles. Without the physical and psychological barriers posed by cars, residents are more incentivised to walk, cycle, or gather in shared spaces. This contributes to healthier urban environments and strengthens the social fabric of neighbourhoods (Bertolini, 2017; Nunen et al., 2011).

Moreover, car-free neighbourhoods contribute to environmental sustainability. The compact design of car-free neighbourhoods reduces the car dependency and energy consumption of the area while supporting mixed-use development with high densities (Ng, 2010).

It also opens up opportunities for climate adaptation, freeing up asphalt space for green spaces and water-permeable surfaces to improve water retention, reduce heat stress, and improve biodiversity (Tang et al., 2019). At the same time, they enhance safety and quality of life by reducing the risk of traffic accidents and creating calmer, more inclusive environments, particularly for children and the elderly (Gallo & Marinelli, 2020).
However, creating a neighbourhood with a complete absence of cars also poses some challenges. The well-connectedness, accessibility, and inclusivity of the neighbourhood could be at risk from the implementation of a car-free system (Basu & Ferreira, 2021; Carvalho et al., 2024; Nielsen et al., 2013; Sun et al., 2017).

Car-Free Neighbourhoods
Residential or mixed-use neighbourhoods that provide a living environment free of traffic, limited to no parking that is separated from residences, and are designed to enable living without owning a car.

3. 2 WELL-CONNECTEDNESS

One of the main challenges of car-free neighbourhoods is that the removal of private cars can unintentionally lead to spatial isolation from the wider city and region (Nieuwenhuijsen et al., 2019). Without the flexibility and reach that cars typically offer, residents may face difficulties in accessing employment, services, and social networks outside the neighbourhood or city (Alessandretti et al., 2023; Bertolini, 2017; Nieuwenhuijsen et al., 2019). If alternative transport modes, such as high-quality public transport and integrated cycling infrastructure, are not properly planned and implemented, the neighbourhood risks becoming disconnected (Alessandretti et al., 2023; Horiike et al., 2023).

The most effective time to implement public transport is at an early stage, before development occurs, as it enhances both inner and outer connectivity within neighbourhoods, presents an attractive and appealing image, and leads to increases in real estate, commercial, and amenity values (Dlabaja, 2023; van der Bijl & Zeeuw, 2009).

For a sustainable car-free neighbourhood, it is therefore important to have other mobility networks that keep the neighbourhood well-connected with the city and region. In practice, this highlights the importance of a continuous cycling network, especially in Amsterdam, where cycling is used to cover long distances (Lonardi et al., 2025; Ton et al., 2017). The growing use of electric bicycles reinforces this need, as they enable bicycle users to travel longer distances (Krukowicz et al., 2021).

Well-Connectedness

The connection of the neighbourhood with other neighbourhoods, the city, and the region through fast, high-quality mobility networks.

3.3 ACCESSIBILITY

Another potential challenge of car-free neighbourhoods is that they may reduce accessibility (Nielsen et al., 2013; Sun et al., 2017). If the internal mobility system is poorly designed, it can result in indirect routes, unsafe crossings, or physical barriers that hinder movement.

This is especially challenging for last-mile trips to homes, amenities, or workplaces (Basu & Ferreira, 2021; Sun et al., 2017). Therefore, a car-free neighbourhood must offer safe, direct, and reliable connections for walking, cycling, and micromobility. This ensures that all residents can easily reach their daily destinations without needing a private car (Marzolla et al., 2024; Nielsen, 2006).

Reggiani et al. (2022) argue that if the number of automobiles in a neighbourhood is reduced, the bikeability and walkability of the neighbourhood should be increased. An increased bikeability entails that the bicycle infrastructure enables and encourages people to access all destinations within and around a neighbourhood (Reggiani et al., 2022). Walkability aims for the same, regarding pedestrians. Factors enhancing walkability and bikeability comprise traffic safety, user comfort, connectivity and directness, dynamic streetscapes, green elements, reduced pollution and cleanliness (Brownrigg-Gleeson et al., 2023; Cabanek et al., 2020).

Besides good quality slow mobility infrastructure, dense and mixed-use land use planning is also essential to keep daily needs accessible without access to cars (Bertolini, 2017). The reach of automobiles is higher than the reach of bicycles or pedestrians; therefore, when planning a car-free neighbourhood, it is essential to design the land use according to these principles, without solely focusing on the infrastructure (Bertolini, 2017; Verkade & Brömmelstroet, 2024).

Accessibility

Ensuring safe, direct, and reliable connections within the neighbourhood for walking, cycling, and micromobility so everyone can easily reach their destination.

3.4 INCLUSIVITY

Mobility should not only be viewed from a technical-efficiency perspective, but interventions must address the needs of individuals from various physically or socially vulnerable groups, e.g. wheelchair users, people with sensory disabilities, pregnant women, and the elderly.

Additionally, transport-related social exclusion affects mostly people with lower income, disabilities, and high age for reasons such as high costs and psychological and physical barriers (Carvalho et al., 2024; Khreis & Nieuwenhuijsen, 2016; Mackett & Thoreau, 2015).

Physical inclusivity is achieved through universal design principles, which involve designing products, environments, programs, and services to be usable by all individuals, as much as possible, without requiring modifications or specialised adaptations (Convention on the Rights of Persons with Disabilities, 2008). Hence, it includes the absence of physical barriers, like steps, barrier kerbs, or cobblestone pavements with large in-fill gaps. It also necessitates shaded paths with greenery, proper maintenance and snow removal, tactile and audible signs operated by hearing-impaired persons, and adequate street lighting (Gupta et al., 2025).

Furthermore, public space is often designed so that it only enables the movement of people, without considering other activities. It should be designed so that pedestrians are enabled to pass through the space and to rest in the space (Gerike et al., 2021). This enables people with reduced walking capacity to also use the public space as their main form of mobility (Stavroulaki & Pont, 2020).

Inclusivity

The mobility network and public space are accessible, safe, and usable for all people, regardless of their physical or social status.

3.5 INTEGRATING WELL-CONNECTEDNESS, ACCESSIBILITY AND INCLUSIVITY IN CAR-FREE DESIGN

To conclude from this literature review, a car-free neighbourhood design potentially causes a reduction in the well-connectedness, accessibility and inclusivity of the neighbourhood. A sustainable car-free design should therefore incorporate mitigating measures to counteract these potential reductions. This can be achieved through design by for example investing in wheelchair friendly public areas with benches for people to rest on and an accessible and reliable public transport network (Alessandretti et al., 2023; Gerike et al., 2021; Gupta et al., 2025; Horiike et al., 2023; Nieuwenhuijsen et al., 2019). Additionally, policy measures could also be considered as mitigating interventions (Gupta et al., 2025; Khreis & Nieuwenhuijsen, 2016). These improvements in well-connectedness, accessibility and inclusivity enable a car-free neighbourhood design on a theoretical level. The connection between these concepts is visualised schematically in Figure 1.

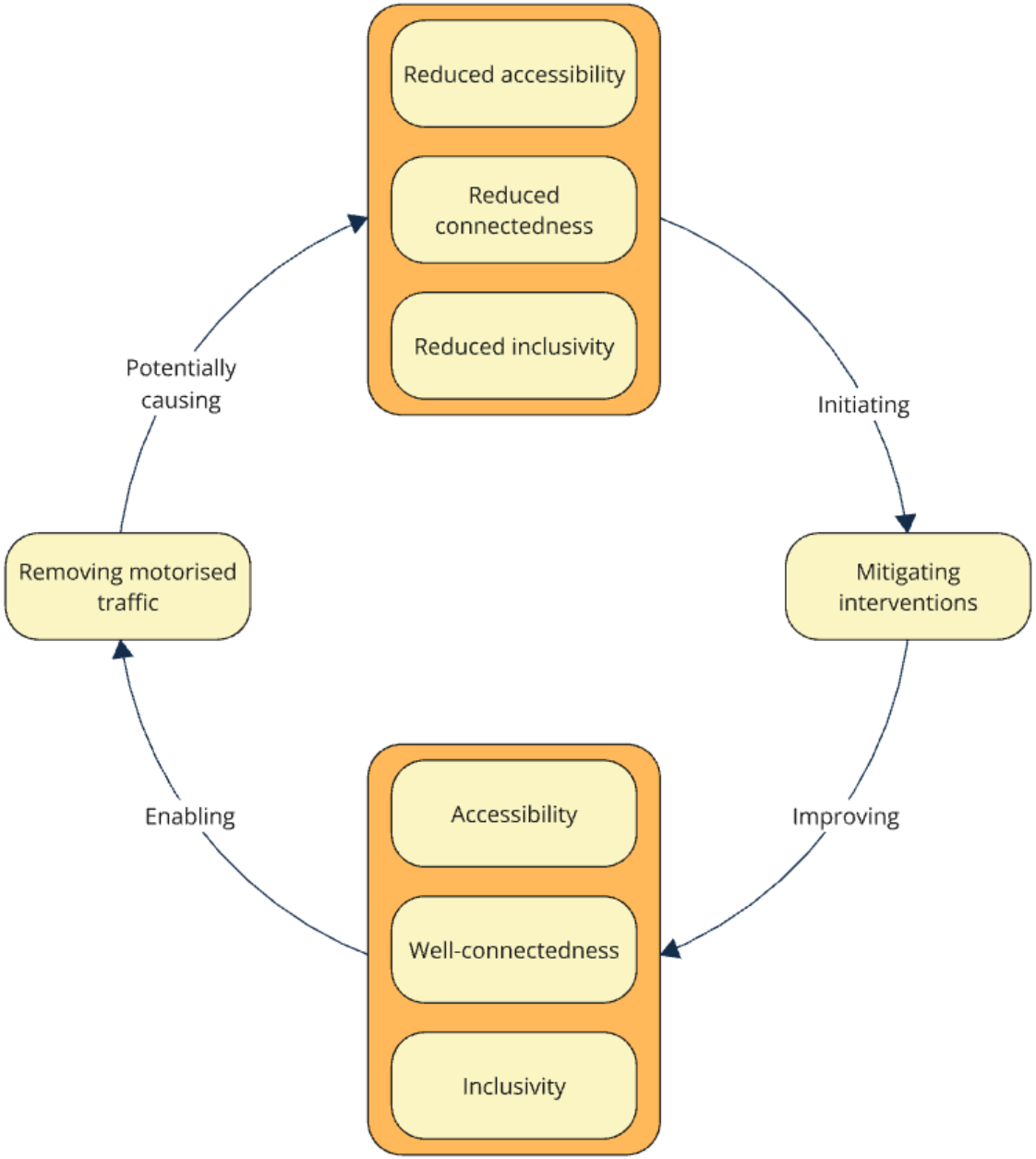


Figure 1: Visualisation of the core concepts of this study and their relations with each other.

4 METHODOLOGY

A mixed-methods approach including both desk and case research was applied in this study, with a schematic overview depicted in Figure 2.

4.1 METHODOLOGY JUSTIFICATION

A mixed-method design combines the depth of qualitative results with the broadness of quantitative analysis, enhancing validity through methodological triangulation (Mackey & Bryfonski, 2018) . Quantitative spatial metrics offer objective measures of accessibility and density, while qualitative interviews and thematic analysis provide contextual insights into stakeholder values and residential experiences (Dehalwar & Sharma, 2024) . This integration strengthens internal and external validity, allowing for the capture of systematic patterns and individual perspectives in the context of the possibility of a car-free Zaanstraat Emplacement.

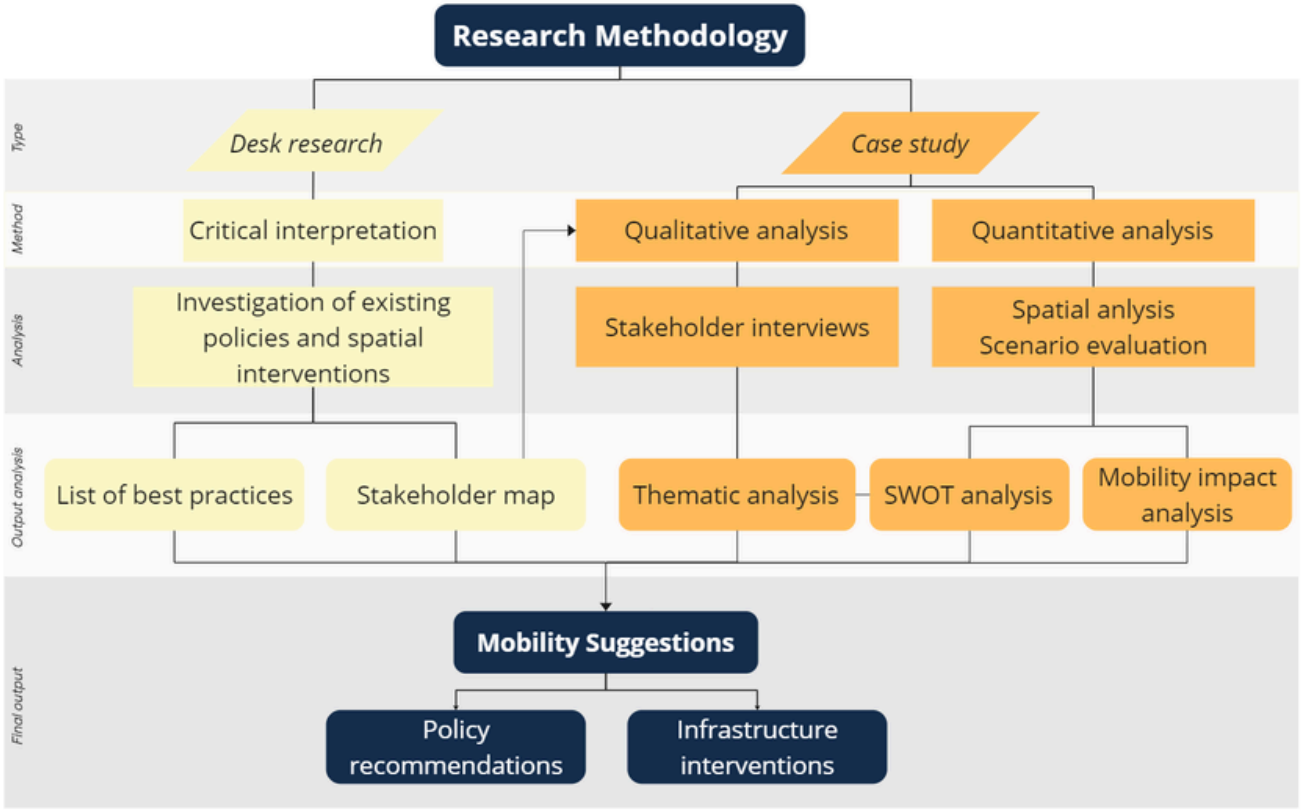


Figure 2: Visualisation of research methodology. The type of research method is colour-coded in yellow for desk research-related methodology workflow and red for case study-related methodology workflow. The different methods, analysis types, analysis outputs, and final outputs, which can be found on the left side of each row, are visualised using different shapes.

4.2 DATA COLLECTION

LITERATURE

A comprehensive desk research was conducted to gather and synthesis secondary data from diverse sources. Key resources include planning documents from the Municipality of Amsterdam on Haven-Stad (Ontwikkelstrategie Haven-Stad, Integraal Raamwerk), and Gemeente Waterleidingen (GWL; one of the first car-free neighbourhoods in Amsterdam). These materials were selected based on their relevance and scope of the research objectives. Additionally, scientific literature was reviewed to identify best reference cases and gain theoretical insights related to, for example, car-free urban environments, mobility planning, and inclusive mobility. The literature review process involved systematically searching databases such as Web of Science and Google Scholar using keywords such as ‘car-free neighborhoods’, ‘urban mobility' and 'car dependency’.

INTERVIEWS

Two formats were used for the triangulation of our data on several aspects.

1. Expert interviews (n=5): Semi-structured interviews of approximately 20-30 minutes with urban planners, mobility specialists, and policymakers were conducted, which aimed to gain more expert knowledge and understand their viewpoints. Whenever possible, these interviews were recorded and subsequently transcribed (n=2).
2. Resident interviews in the car-free neighbourhood of GWL (n=7): Additionally, semi-structured interviews of 10-15 minutes were conducted with GWL residents that sought to explore their experiences and perceptions of living in a car-free neighbourhood. These interviews were recorded or summarised (see Appendix 3).

All participants provided informed consent. Anonymity and data protection complied with institutional guidelines. The recordings and notes are solely used for this research and shared in aggregate form with the municipality of Amsterdam.

SPATIAL DATA

The currently existing road, cycling, pedestrian, and public transport network of Haven-Stad was sourced from OpenStreetMap, which provides various kinds of open-source geospatial data (OpenStreetMap, n.d.) .

Since Haven-Stad’s comprehensive mobility network is set to significantly change with its redevelopment, future infrastructural interventions were obtained from the Integraal Raamwerk policy document, which shows the future bicycle, pedestrian, and public transport networks (Gemeente Amsterdam, 2021b) . Any newly proposed road segments were manually added to the existing network map sourced from OpenStreetMaps, yielding a future road network map of Haven-Stad. Additionally, the individual neighbourhood boundaries of Haven-Stad were manually drawn, based on the data provided by Ontwikkelstrategie Haven-Stad (Gemeente Amsterdam, 2017) . This document also provided several function programma data, such as the projected number of workplaces and dwellings in each neighbourhood. The neighbourhood data, together with the future road network, serve as inputs for the Car Dependency Index analysis.

4.3 DATA ANALYSIS

DESK RESEARCH: STAKEHOLDER ANALYSIS AND LISTING REFERENCE CASES

Global reference cases that are similar regarding urban mobility challenges were compiled (Chapter 5.3), along with a mapping of the stakeholders (see Appendix 2) that are involved in the Zaanstraat Emplacement, specifically mobility-related actors. The latter was carried out based on the model proposed by Prados et al. (2022). The reference cases were chosen based on their implementation of key car-free development strategies, the availability of practical and policy-based interventions, and their potential to offer transferable insights despite differences in scales or local context compared to Zaanstraat Emplacement. The stakeholder map aims to categorise relevant participants based on their institutional functions and relationships, which helps identify the interdependencies and coordination needs for the inclusive and effective implementation of a car-free Zaanstraat Emplacement.

THEMATIC ANALYSIS

To help understand the transcribed semi-structured interviews conducted at GWL-terrein, a thematic analysis was utilised to simplify the outcomes of the interviews.

By coding the recorded transcripts into different codes and code groups, the data became more manageable and usable for the SWOT analysis. Coding was carried out manually with the assistance of ATLAS.ti, a software programme commonly used in qualitative data analysis. Using an inductive approach, codes were generated based on both what residents mentioned during the interviews as well as theoretical concepts emerging from the theoretical framework. The data was coded in a way that allowed us to easily coordinate through the data and find quotes and opinions of residents that support our SWOT analysis.

SWOT ANALYSIS

For this research, the most commonly used definitions of the four SWOT components were modified to better suit the case scenario analysis. SWOT analyses are commonly used as a strategic management tool to identify competitive advantages and drawbacks, which is more flexible than scenario planning (Luoto & Lonkila, 2018).

Strengths and weaknesses, which make up the first two letters of the SWOT acronym, are considered internal factors, which are the resource capabilities and deficiencies. In contrast, opportunities and threats are frequently defined as external factors related to their environment (Takahashi & Maeno, 2011).

In this case, a SWOT analysis of the Zaanstraat Emplacement neighbourhood is conducted from a mobility perspective, identifying factors relevant to implementing a car-free neighbourhood design. A scenario planning-based SWOT analysis method inspired by Luoto and Lonkila's (2018) method was chosen, where the SWOT analysis focuses on a future scenario and identifies key drivers and factors. The key drivers and factors are based on the conducted interviews and expert knowledge, which are validated with scientific literature. Furthermore, we choose to redefine the internal and external factors to specifically explain the (pre-)conditions, which can be described as factors (strength and opportunities) that must be utilised to enable a car-free neighbourhood, and barriers, which can be described as factors (weaknesses and threats) that impede the transition towards a car-free neighbourhood.

MOBILITY IMPACT ANALYSIS

A mobility impact analysis (MIA) was conducted to identify the possibilities of the area regarding its mobility network, and the potential risks, drawbacks of the future mobility network (e.g. missing public transport connection at a certain area).

It consists of two separate analyses. Firstly, a Car Dependency Index (CDI) was developed for each neighbourhood in Haven-Stad, which combines three different metrics: active mobility accessibility of dwellings and workplaces, density, and function/land-use mix (see Figure 3). The accessibility modelling was based on planned bicycle and pedestrian networks as outlined in existing policy documents, along with projected programme data on the number of dwellings and workplaces expected in each neighbourhood. Secondly, a Multi-Criteria Analysis (MCA) was conducted to identify the most suitable public transport network for the mobility hub. Since public transport is a key factor in creating car-free living environments, we aim to systematically evaluate the four metro expansion scenarios proposed in the Integraal Raamwerk policy document (Gemeente Amsterdam, 2021b), along with two scenarios that do not include any metro expansions.

1. *Car Dependency Index (CDI)*

The CDI is constructed as a normalised composite index, where each neighbourhood receives a score between 0 and 1. A score of 0 represents the lowest risk of car dependency as a result of high accessibility to destinations (dwellings and workplaces) by walking or cycling, high density, and a balanced mix of land use functions. By assigning a numerical score to each neighbourhood, it becomes possible to compare and rank them against each other in terms of expected continued dependency on cars, and to assess the suitability of Zaanstraat Emplacement for implementing a car-free approach.
A low score would indicate that it is less prone to continued reliance on car usage, which would mean Zaanstraat Emplacement has favourable built environment factors contributing to a successful implementation of a car-free approach.

The CDI relies on three metrics, expressed in Equation 4.1:

$$\begin{aligned}\text{CDI} = \; & 0,7 \cdot (\text{Accessibility}_{\text{workplaces}} + \text{Accessibility}_{\text{dwellings}}) \\ & + 0,2 \cdot \text{Density} \\ & + 0,1 \cdot \text{Land-Use Mix}\end{aligned}$$

The destination accessibility metric comprises dwellings and workspaces. The index is calculated by adding the sum of each metric multiplied by its respective weight. These metrics and weights were based on the findings of Ding et al. (2017) , who found that 'the factors of connectivity and accessibility are found to have significant negative effects on car ownership', whereas land function mix, while it can contribute to a lower tendency to own cars, its effect is not as strong as a highly accessible neighbourhood. Therefore, a weight of 70% was assigned to the first metric, per workplace/dwelling accessibility. Consequently, while high urban densities have been shown to correlate with lower car usage, the relationship is more ambiguous than for accessibility (Kim & Brownstone, 2013; Schimek, 1996) . Therefore, it gets assigned a weight of 20%. Lastly, while land use mix (10%) has also been associated with reductions in car usage, its effect, as shown by Ding et al. (2017) , is not as strong as the other metrics mentioned.

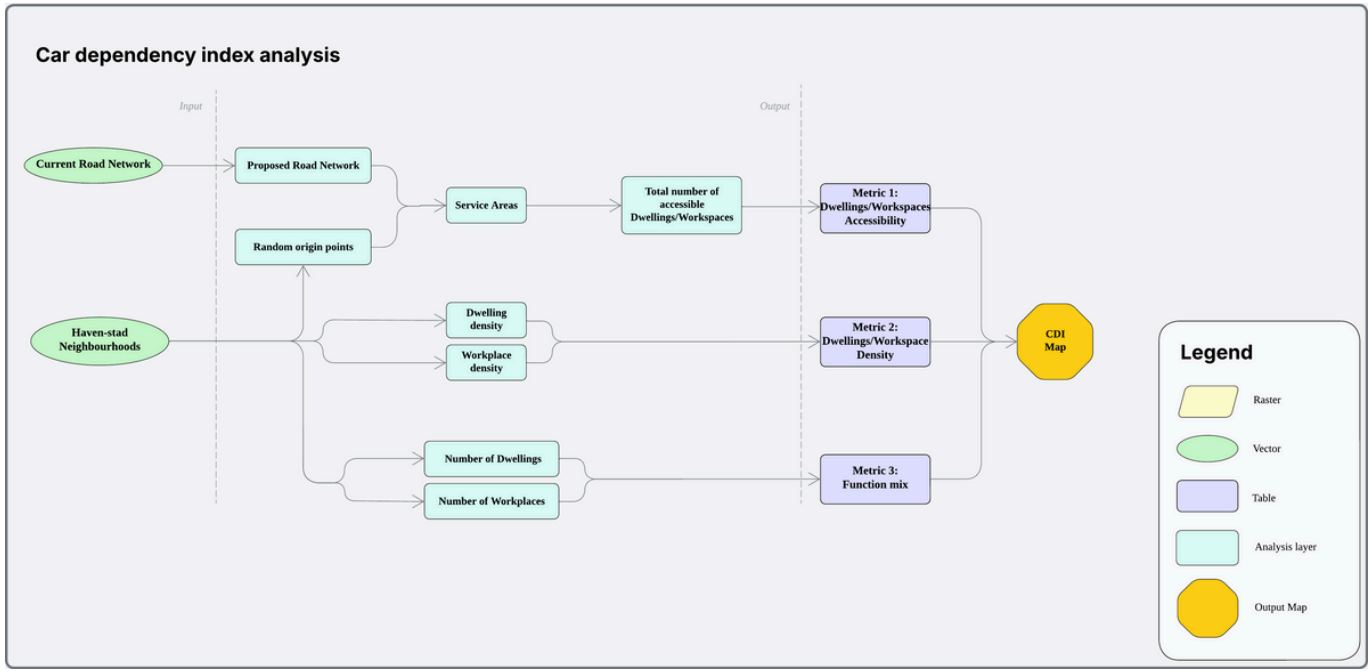


Figure 3: This figure shows the workflow that was utilised to generate the Car Dependency Index for each neighbourhood in Haven-Stad. An in-depth explanation and justification of each metric and its weight can be found in Appendix 5.1.

2. Multi-criteria analysis of public transport networks

Six distinct public transport scenarios were defined and compared using a multi-criteria analysis to identify the optimal, future-proof public transport network. During the analysis, each scenario was assessed on a scale of 1 to 10 based on the specified criteria.

Each evaluation criterion was weighted, taking into account a mobility-focused approach. The final score for each variant is derived from the weighted sum of these two numbers.

There were eight criteria in total identified for evaluating the scenarios, some of which are directly linked to mobility, while others are indirectly connected, such as greenery. The criteria for the MCA include the number of dwellings and workplaces surrounding the rapid rail transit stations, the functional coverage ratio (developed by us), the theoretical maximum capacity of the rail transit, the degree of intermodality, the potential for connecting pedestrian and cycling networks to the stations, the management of these two modes, the impact on greenery, and the financial aspect (see Figure 4). The assessed indicators are considered widely used in transport studies, but only those that could be evaluated using the available data and evaluation tools. A more detailed description of the indicators can be found in Appendix 5.2. The MCA considered not only the demands of Zaanstraat Emplacement but also the entire Haven-Stad.

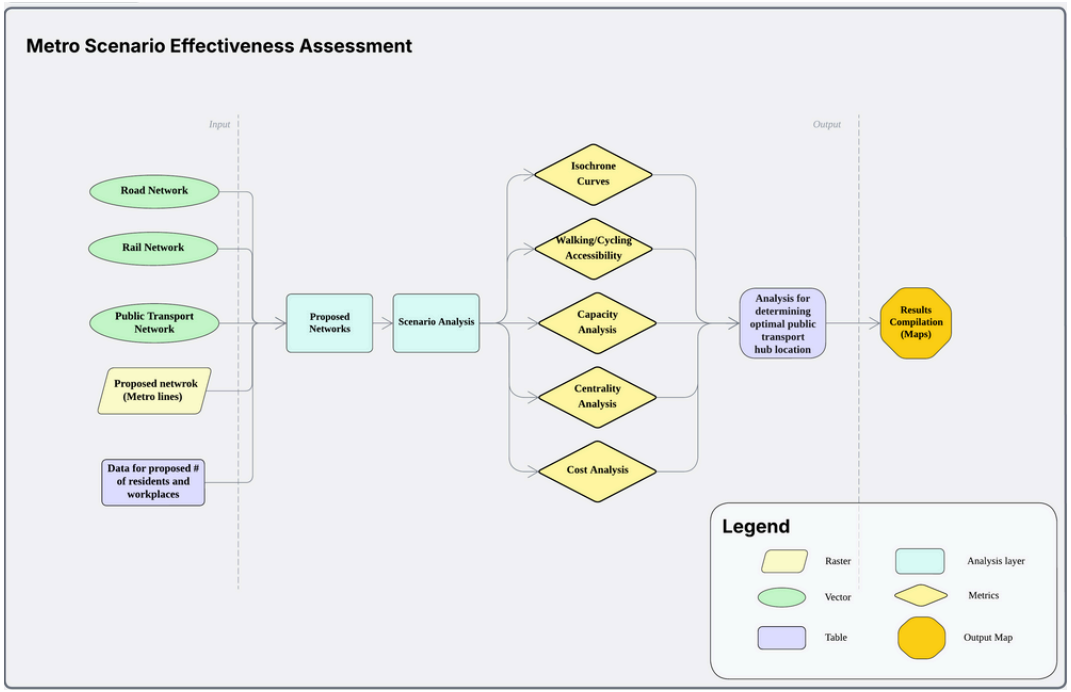


Figure 4: This figure shows the workflow that was utilised to create the MCA analysis. An in-depth explanation and justification of each metric and its weight can be found in Appendix 1.6.

The weighting of the scores was established from a mobility perspective, meaning that those criteria directly related to mobility have a higher weight. Greater weight (0.15) was given to the accessibility of residential and business land uses, emphasising the importance of situating stations in areas fully surrounded by built-up land to ensure optimal connectivity for most users. Connection to the pedestrian network was also highly weighted (0.15), reflecting the importance of station accessibility through the development of a slow mobility network that allows direct routes to hubs without unnecessary detours. The highest weight was allocated to cost (0.20), to ensure that the project remained financially feasible. The MCA considered not only the demands of Zaanstraat Emplacement but also the entire Haven-Stad, as public transport has a broader impact on the network (Geurs & van Wee, 2004) .

5 ANALYSIS

This chapter examines key findings to implement a car-free neighbourhood in Zaanstraat Emplacement. Chapter 5 is structured into three subchapters. First, the conditions and barriers to implementing a car-free neighbourhood are discussed. Second, the spatial mobility design to support such a neighbourhood is discussed. Finally, policy interventions to reduce car dependency among residents and reference cases of car-free neighbourhoods are discussed.

5.1 (PRE-)CONDITIONS AND BARRIERS TO IMPLEMENTING A CAR-FREE ZAANSTRAAT EMPLACEMENT

This section presents a synthesised SWOT analysis (Figure 5) with a reframe of the SWOT categories, in which the SWOT analysis focuses on a future scenario (Car-free Zaanstraat Emplacement). The Strengths and Opportunities can be seen as the enabling conditions whereas the Weaknesses and Threats can be seen as the barriers that impede the implementation of a car-free Zaanstraat Emplacement.

Figure 5: SWOT analysis considering a future car-free Zaanstraat Emplacement scenario. The elaborated version of the SWOT analysis, with additional explanations and scientific references, can be found in Appendix 4.

ENABLING CONDITIONS

Four enabling conditions based on the Strengths and Opportunities were found in the SWOT analysis. Together, these conditions form a prerequisite foundation to successfully integrate car-free design in Zaanstraat Emplacement.

One of the key internal strengths of a possible car-free Zaanstraat scenario is the more sustainable living environment, which aligns with scientific literature from the theoretical framework, which suggested that car-free environments create more space for greenery, reduce emissions, and become safer.

Multiple interviewees from GWL-terrein acknowledge these benefits of living at GWL-terrein. Additionally, they stated that GWL-terrein is a little village inside the city of Amsterdam, which makes them feel very connected to their neighbours as they know each other well and organise events. This sentiment of strong social cohesion and happiness is backed up by Amsterdam's city survey results, from which it can also be concluded that the GWL-terrein neighbourhood consistently ranks at the top of the municipality's happiness survey (GWL Terrein, 2024).

The most mentioned precondition by the interviewees from GWL-terrein for a successful transition to a car-free neighbourhood is establishing a dense and reliable public transport network surrounding the neighbourhood, making it well connected with major train stations. Although the Zaanstraat Emplacement area lies between Station Sloterdijk and Amsterdam Central Station and thus has an advantageous strategic location, the accessibility of the network is what counts. Insights from the GWL-terrein interviews revealed that these residents are most concerned about accessible public transport, especially in times of construction work on roads, public transport lines, and strikes.

Secondly, shared mobility hubs for cars, bicycles and other innovative transportation modes could be a supporting precondition to bridge the gaps between fixed transit routes and addressing last-mile connectivity (M. Walvius, personal communication, June 11, 2025). Interviewees from GWL-terrein identified the opportunity to leverage shared car cooperatives, micro-transit services, and to place a sufficient number of hubs near the edges of the car-free area. Research substantiates this point as it shows that offering multiple modes of transport and building necessary active mobility infrastructure could lead to a reduction of car dependency among urban residents, thus counteracting car dependency factors (Basu & Ferreira, 2021; Nielsen et al., 2013; Sun et al., 2017).

Thirdly, prioritising active mobility can constitute an indispensable precondition. Given that Zaanstraat Emplacement does not have dwellings and workplaces yet, there are opportunities to design the space with encouraging active mobility in mind and simultaneously reducing car dependency (W. Greeve, personal communication, June 6, 2025).

Finally, the SWOT analysis outcome was discussed with S. Elfferding in comparison with Hamburg's new car-free neighbourhood at Kleiner Grasbrook. Efferding mentioned that extending the metro network is a fundamental criterion, along with improving ferry services and cycling routes leading to the area. It was emphasised that there are already existing car-free areas in Hamburg that meet the needs of residents, and more neighbourhoods will be redeveloped in this manner (S. Elfferding, personal communication, June 27, 2025) .

BARRIERS

Despite these enabling conditions, some barriers, based on the Weaknesses and Threats, were found during the analysis that may impede a car-free Zaanstraat Emplacement.

A key weakness of transitioning towards a car-free Zaanstraat is the car dependency of people and the risk of exclusion. Insights from the interviews at GWL-terrein revealed that some residents simply need a car to get to either do or go to their jobs. There are many variables that have an effect on car dependency, such as built environment, socio-economic factors, car ownership, lifestyle, and public transport, which makes the willingness of people to live in a car-free neighbourhood a difficult factor to determine beforehand (Saeidizand et al., 2022). Translating this to a barrier could mean that the unpredictable nature of car d ependency among re sidents is a main barrier . Unlike infrastructure, which can be planned and executed fairly easily , individual attachment to private vehicles and , with that, their behaviour and sentiment towards car-free neighbourhoods are more difficult to know. Amongst the GWL interviewees, the sentiment towards living in or close to a car-free neighbourhood differ ed greatly from people stating that they chose the car-free lifestyle to people stating that a car is a must-have. During the interviews at GWL-terrein, logistics challenges were mentioned as well. To receive packages, grocery deliveries, or move furniture, it takes more effort since parking is only allowed at the border of the GWL-terrein, which necessitates using a trolley to deliver /move products.

Another barrier brought up by one of the interviewees is the rigidity of existing parking legislation. Current guidelines from CROW, a think tank specialised in built environment policymaking, advise a certain number of parking spaces per housing unit, the so-called parking norm. It occasionally occurs that residents sue developers or municipalities demanding more parking spaces than municipal policies prescribe, which can lead to project delays and higher costs (S. Drolsbach, personal communication, June 6, 2025).

A third barrier involves the vulnerability of the public transport system itself. Several interviewees of GWL-terrein indicated that their car-free neighbourhood was not as accessible as it should be due to construction work on the public transport lines surrounding their neighbourhood. This forms a challenge, because many interviewees mentioned relying on public transport as their neighbourhood is car-free.

5.2 ANALYSING THE IMPACT OF MOBILITY NETWORKS IN ZAANSTRAAT EMPLACEMENT

In order to support car-free living, alternative mobility modes should serve as feasible substitutes. Therefore, this chapter consists of two infrastructure- and spatial-focused analyses that were conducted within the scope of Zaanstraat Emplacement and Haven-Stad. These analyses aim to assess the planned infrastructure in and around the Zaanstraat Emplacement and its suitability in supporting walking, cycling, and public transport as primary modes of mobility, which makes up the second sub-research question.

ASSESSING THE SUITABILITY OF ZAANSTRAAT EMPLACEMENT FOR A CAR-FREE APPROACH USING THE CDI

Before reviewing the final car dependency score of Zaanstraat Emplacement, the intermediate metric scores will be discussed. The first metric that makes up the CDI is the 'per workplace/dwelling accessibility'. The numeric values in the tables below represent the average number of dwellings or workplaces accessible in Zaanstraat Emplacement for 5 minutes of cycling or 15 minutes of walking.

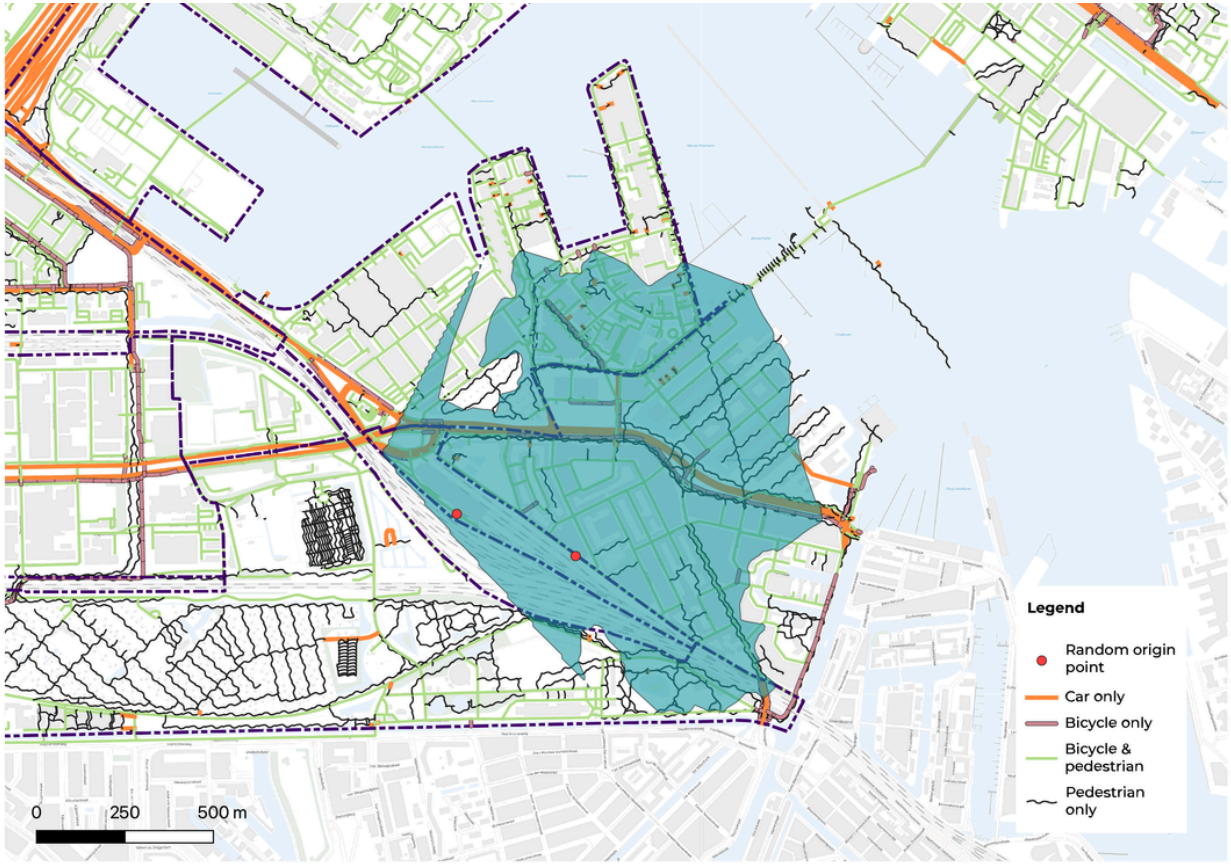
Neighbourhood	Bike, dwellings, 5 minutes	Walking, workplaces, 15 minutes
Alfadriehoek	4518	1681
Coen- en Vlothaven	2756	1824
Cornelis Douwes 0-1	2698	2657
Cornelis Douwes 2-3	1747	1703
Melkweg Oostzanerwerf	407	154
Minervahaven	2447	2395
Sloterdijk Centrum / Centrum-Noord	1423	1228
Sloterdijk I Noord en Zuid	3592	3500
Sportpark Transformatorweg en Amsterbaken	9889	7416
Zaanstraat Emplacement	**7502**	**1885**

Table 1: Average number of workplaces accessible per neighbourhood (Per Workplace Accessibility)

Neighbourhood	Bike, dwellings, 5 minutes	Walking, dwellings, 15 minutes
Alfadriehoek	5963	4684
Coen- en Vlothaven	2992	2248
Cornelis Douwes 0-1	3002	3017
Cornelis Douwes 2-3	1879	1798
Melkweg Oostzanerwerf	5296	4812
Minervahaven	2752	2577
Sloterdijk Centrum / Centrum-Noord	2465	2044
Sloterdijk I Noord en Zuid	2776	2520
Sportpark Transformatorweg en Amsterbaken	12707	10833
Zaanstraat Emplacement	**8954**	**3310**

Table 2: Average number of dwellings accessible per neighbourhood (Per Dwelling Accessibility)

When reviewing the Workplace and Dwelling Accessibility for Zaanstraat Emplacement (Table 1 and 2), it becomes apparent that the accessibility when considering bicycles ranks second highest. Factors contributing to this result are its central location within the southern half of Haven-Stad and the fact that the majority of Minervahaven and Sloterdijk I Noord en Zuid can be reached, which will both be relatively workplace-dense areas.
While Zaanstraat Emplacement scores high on projected workplace and dwelling density, it also faces significant accessibility barriers that could negatively impact the attractiveness and efficiency of active mobility.



Base map: PDOK, road network: OSM, neighbourhood boundaries: Ontwikkelvisie Haven-Stad 2017

Map 2: Service area of Zaanstraat Emplacement, dwellings, 15 minutes of walking.

The area is bounded by a railway yard and rail line to the south, which physically separates it from Westerpark. Currently, there are only two points of access across this barrier: a pedestrian/bicycle tunnel at the eastern edge of the neighbourhood and a secondary crossing at the far western boundary. No planned additional crossings were identified in the policy documents on Haven-Stad.

The resulting limited permeability that forces traffic users to cross the barrier through 2 access points is not strongly reflected in the bicycle accessibility analysis for workplaces and dwellings, as the higher speed means even a small 5-minute time window would cover a relatively large area. The barrier effect that the railway line and yard pose becomes significantly more pronounced when considering walking, as slower walking speeds result in a smaller area covered and could potentially not reach crucial access points across the barrier, such as the tunnel to Westerpark in the eastern part of Zaanstraat Emplacement. Map 2 shows the area accessible from 2 random origin points in the neighbourhood within 15 minutes of walking.

The second metric that contributes (20%) to the calculation of the Car Dependency Index is workplace and dwelling density. Zaanstraat Emplacement has the second-highest projected workplace density behind Sloterdijk Centrum/Centrum-Noord and the highest projected dwelling density.

Lastly, the third metric is the land use function mix (10% weight), which evaluates the balance between residential and working functions within each neighbourhood.

Across most neighbourhoods in Haven-Stad, except for Sloterdijk Centrum/Centrum-Noord, the projected land use is mostly residentia l. On average, approximately 66% of the total land use is allocated to residential functions. This relatively uniform distribution of land-use shares throughout Haven-Stad means that Zaanstraat Emplacement does not significantly stand out in terms of functional diversity.

Neighbourhood	CDI
Sportpark Transformatorweg en Amsterbaken	0.05
Zaanstraat Emplacement	**0.40**
Alfadriehoek	0.57
Sloterdijk I Noord en Zuid	0.60
Minervahaven	0.65
Cornelis Douwes 0-1	0.66
Coen- en Vlothaven	0.67
Cornelis Douwes 2-3	0.70
Sloterdijk Centrum / Centrum-Noord	0.77
Melkweg Oostzanerwerf	0.93

Table 3: Car Dependency Index (CDI) of Haven-Stad neighbourhoods.

Combining the three metrics: active mobility accessibility, density, and land-use function mix yields a Car Dependency Index score of 0,40 for Zaanstraat Emplacement, making it the second-lowest scoring neighbourhood in terms of CDI in Haven-Stad (Table 3). Since a lower score represents a lower risk of car dependency, the result suggests that a relatively strong foundation for car-free living will be realised with the proposed plans for Zaanstraat Emplacement. The neighbourhood's favourable CDI is primarily driven by its high per workplace/dwelling accessibility, which makes up the most heavily weighted component of the index (70%). This number indicates that, on average, a large number of destinations can be reached from each dwelling or workplace within n minutes of walking or cycling, which is an essential precondition for fostering car-free urban living (Khreis & Nieuwenhuijsen, 2016).

MOBILITY NETWORK IDENTIFIED BY MCA

The previous chapter identified Zaanstraat Emplacement as one of the neighbourhoods with built environment factors that make it favourable for implementing a car-free area. This assessment was based on three key factors: accessibility by active modes, residential and workplace density, and land-use function mix. However, in addition to these built environment factors, the availability and quality of public transport also play a crucial role in supporting car-free living. To examine this component, the future public transport network was analysed based on 6 scenarios. The future public transport network is crucial for the area, as it ensures connectivity and therefore has a significant impact on the entire mobility system.

Scenario I represents the extension of the metro track in an elevated manner, reaching Centraal Station without closing the metro loop, despite financial and technical difficulties. Scenario II employs a cut-and-cover tunnelling technique with a s lightly different route than Scenario I, providing slightly greater coverage for residential and business areas.

Scenario III features a more innovative route for Amsterdam, as the metro does not run alongside the train tracks, encompassing new areas within the network, albeit in a cost-intensive manner. Scenario IV adds a direct regional connection to Scenario III with the train station, resulting in a highly developed network. Scenario V indicates the absence of metro expansion, whereby train services can partially fulfil their role. Scenario VI explores the possibility of serving Haven-Stad solely by trams, which is not a unique solution for Amsterdam (IJburg). All scenarios are considered technically feasible, but with a broad range of financial requirements (K. Ruchtie & R. Sutmuller, personal communication, June 12, 2025) . A detailed description of the Scenarios, including a map visualisation, can be found in Appendix 5.2.

The analysis (Table 4) reveals that all scenarios provide advantages in various aspects, making the outcome largely dependent on the wei ghts as signed to the indicators. The scenarios scored within a range of 1 point on a 1 to 10 scale, indicating that there is no significant difference in their feasibility.

However, Scenario III, which includes the metro connection through Houthavens, received the highest score. This is due to its route being closer to the geometric centre of the Haven-Stad area, better connectivity to the walking and cycling networks, and minimal impact on green spaces. Still, it is important to note that this scenario is very costly because of its construction method. The analysis also indicates that extending public transport has great potential to support the car-free design of the Zaanstraat Emplacement neighbourhood.

Table 4: Results of the MCA analysis

Scenarios	Weight	I Metro alongside the train track	II Metro mainly alongside the train track, but with a station closer to Havenstad	III Metro through Houthavens	IV Metro through Houthavens and train station at Hemknoop	V No metro extension, train station at Hemknoop	VI Only tram lines
Accessibility of residential and business areas	0,15	4	5	7	9	2	0
Functional coverage ratio	0,09	5	6	7	7	5	8
Capacity	0,08	8	8	8	9	5	3
Intermodality	0,10	3	5	8	7	2	5
Connection to the pedestrian network	0,15	6	7	9	9	6	10
Connection to the cycling network	0,10	8	8	8	8	9	4
Greenery	0,08	4	7	9	8	10	8
Cost	0,25	6	4	2	1	7	9
MCA score		**5,51**	**5,84**	**6,49**	**6,44**	**5,70**	**6,25**

5.3 INTERVENTIONS TO CREATE CAR-FREE NEIGHBOURHOODS

This section addresses the final research question: 'What policy and governance solutions can be implemented to ensure that residents can live/work in Zaanstraat Emplacement without relying on cars?'. Based on national and international references study, policy documents, and field study of i.e. King's Cross (London), it outlines physical and non-physical strategies that together contribute to a viable car-free future in Zaanstraat Emplacement.

PHYSICAL AND TANGIBLE INTERVENTIONS

This section explores physical design interventions, such as mixed-use development, mobility hubs, and high-quality pedestrian and cycling infrastructure, which can be critical to supporting car-free living in Zaanstraat Emplacement.

a. Mixed-Use Development

Mixed-use zoning, which combines residential, workplace, retail, service, and entertainment facilities within walking or cycling distance, is one of the main strategies for building a car-free neighbourhood (Arup Australia Pty Ltd., 2025; Iannillo & Fasolino, 2021) . It aligns with the principles of the 15-minute city, where essential services are accessible within a short walking or cycling distance rather than relying on cars (Allam et al, 2024). A car-free neighbourhood in the USA, Culdesac Tempe, has become a successful mixed-use development case based on its unique car-free design concept and high-density land use (Caso, 2023; Parolek, 2025; Steuteville, 2024; WSPP, 2025).

King's Cross, situated in London, the United Kingdom, demonstrates how this principle functions in practice. The area features 1700 private residences with both social and affordable housing, two primary schools, shaded public spaces, ground-floor retail stores, etc., integrated into former railway structure (A. Harries, personal communication, June 17, 2025). Shared facilities promote community interaction, and with most daily users residing within a 2-mile radius, walking and cycling become the primary modes of transportation. This spatial logic can be directly applied to Zaanstraat Emplacement, encouraging dense, mixed-use development that reduces future reliance on cars and enhances local vitality.

b. Public Transport Integration

Establishing a multi-model transportation hub that integrates metro, bus, and cycling infrastructures can significantly enhance the efficiency and attractiveness for people to use public transportation (Mishra, 2020; Quiroz Villanueva et al., 2025). King's Cross is located beside one of Europe's largest transportation hubs, London King's Cross Station. The station design ensures that pedestrians can enter and exit public buildings from multiple directions and have straightforward neighbourhood and community routes (A. Harries, personal communication, June 17, 2025). It is important to optimise the layout of public transportation routes and stations according to the travel needs of community residents, increase the coverage of bus routes, improve the density of stations, and enable residents to use public transportation more conveniently (Heyken Soares, 2021). This permeability improves accessibility and encourages people to use public transportation. Zaanstraat Emplacement should also prioritise high-quality intermodal transport links and design user-friendly pedestrian routes.

c. Parking and Urban Logistics

Effective parking management and logistics planning are crucial to maintaining low-traffic or car-free zones. Research suggests that restricting parking supply and discouraging vehicle access can significantly reduce car use (Barter, 2022; Rye et al., 2023). At King's Cross, there are several strategies that Zaanstraat could also utilise:

1. Residential parking is limited to underground spaces, and commercial units have no dedicated parking.
2. Retailers must schedule deliveries between 7:00-10:00 AM, which is a strict rule sign in their lease agreements. Missing a time slot means that delivery will not be possible on that day.
3. Additionally, parking enforcement is visible and rigorous. A UK parking Control LTD sign mandates:

(i) Active loading/unloading only in designated areas.
(ii) Maximum stay time is 20 minutes, and no return within 90 minutes.
(iii) No parking in yellow lines or shaded areas. (iv) Fines of £100 are enforced 24/7 via CCTV and ANPR systems.

This structure reduces car activity while ensuring access for essential services.

It also highlights the importance of embedding logistics rules into legal agreements and urban design (A. Harries, personal communication, June 17, 2025). Zaanstraat could adopt similar time-limited delivery slots, edge parking, and signage-based enforcement measures to proactively regulate vehicle access.

Meanwhile, the strategic utilization of retractable barriers or timed gates, as well as infrastructure build in Amsterdam's Frans Halbuurt car-free zone, has proven effective in preventing private cars access while maintaining emergence and necessary access (Foletta, 2011; Gemeente Amsterdam, n.d.; Melia, 2014). Thus, implementing such access control infrastructure at key entry points in Zaanstraat can help to build a pedestrian friendly area where cars are prohibited from entering and parking.

d. Pedestrian and Cycling infrastructure

Building a coherent network of pedestrians and bicycle paths to allow for the convenient and safe travel of pedestrians and cyclists in the community and surrounding areas is important to reduce car usage (Steinacker et al., 2022; Wendling et al., 2020). In Zaanstraat Emplacement, connecting pedestrian and cycling routes directly to housing, amenities, and public transport is expected to improve daily mobility model.

This can reduce dependency on cars by potential residents and workers in the Zaanstraat area. Meanwhile, it is also important to improve the supporting facilities of pedestrians and bicycle lanes, such as lighting facilities, rest areas, drinking water points, etc., to improve the living experience in car-free contexts (Argent LLP, 2021; Equipaje, 2018; Frank et al., 2015).

In the King's Cross area, pedestrians benefit from shaded walkways, rest areas, drinking fountains, and visual guidance such as painted roads and signs (A. Harries, personal communication, June 17, 2025; Tufail et al., 2021). It is worth noting that elderly residents reported no major accessibility issues living at King's Cross, and while there is a lack of private parking spaces, the majority still actively choose to live in this car-restricted area (A. Harries, personal communication, June 17, 2025). These features demonstrate that car-free communities can remain inclusive when designed following general accessibility principles. Zaanstraat Emplacement would also benefit from applying similar design standards to ensure comfort, safety, and usability in the community.

NON-PHYSICAL MEASURES

Tangible spatial interventions play a role in making Zaanstraat Emplacement liveable for car-free residents; however, non-physical spatial factors could potentially provide additional benefits. These are presented below.

a. Governance and Policy Enforcements

To sustain car-free living, governance funding and regulations must contribute to facilitating car-free design principles. The example of the GWL-terrein neighbourhood in Amsterdam indicates that such agreements can effectively regulate car access and parking behaviour, thereby strengthening the integrity and functionality of car-free communities (Kushner, 2005; McAskie, 2021). Coincidentally, the car-free neighbourhood mentioned earlier, Culdesac Tempe (US), also requires residents to sign a contract that bans them from parking their car within 400 metres of the neighbourhood's core area to further enhance the goal of a car-free community (WSPP, 2025). The community is specifically designed without private vehicle infrastructure, while prioritising walking, cycling, and shared electric cars. This enforceable legal provision is a residential condition that plays a central role in stimulating a car-free lifestyle.

b. Engagement and Communication

Activities that induce a behaviour change, such as public education, car-free days, and participatory events, have been shown to play a key role in building public acceptance of car-free lifestyles. For example, the city of Oslo, Norway, promotes the concept and advantages of a car-free life to residents by organising car-free day activities carrying out publicity and education, and guiding residents to reduce the use of private cars. Its advantage is to improve residents' awareness and participation in environmental protection, create a good car-free living atmosphere, and promote the long-term development of car-free communities (Rydningen, 2017). The campaign and activities employed in Oslo could serve as an example of a non-physical measure that can be employed in fostering a car-free neighbourhood design in Zaanstraat Emplacement.

At King's Cross, interviews with retail staff highlighted the impact of parking restrictions on daily activities. Couriers use trolleys for deliveries, and due to the lack of direct parking spaces, delivery drivers often struggle to locate stores (A. Harries, personal communication, June 17, 2025).

These findings highlight the necessity of robust communication strategies, onboarding training for new tenants, and improved navigation systems. Zaanstraat Emplacement should initiate participatory planning, car-free education campaigns, and signage-based guidance to ensure a smooth implementation of the car-free principles.

6 DISCUSSION

This study used a mixed-methods approach combining a mobility impact analysis (focused on spatial data evaluation) with interviews conducted among 2 target groups: relevant field experts and residents from a currently existing car-free neighbourhood, GWL-terrein. To better organise and interpret the interview transcripts, the qualitative data was analysed using a thematic and SWOT analysis.

From the interviews, it emerged that all participants living in the car-free GWL-terrein neighbourhood express high satisfaction with the car-free nature of their living environment. Several participants highlighted that the absence of cars creates a more pleasant environment, especially for families with young children. As Participant 2 explained:

> *"I have two daughters aged 8 and 10 and they can easily play and cycle around here, this was absolutely a deliberate choice why I chose to move to this neighbourhood."*

Another participant mentioned the lack of car noise :

> *"It is just quiet here, I love the peace and quiet."*

These findings corroborate existing literature linking cars in living environments to negative externalities. Numerous studies have shown that car traffic contributes to noise pollution, air pollution, and accidents, the latter of which can contribute to a sense of insecurity for groups such as children (Hensher, 2008; Magdin et al., 2019). In an environment free from cars, such as GWL-terrein, these externalities are greatly less pronounced or absent, enhancing the perceived quality of life of residents, which is also reflected in the interviews.

However, when the topic of accessibility was discussed, some participants mentioned that public transport in and around GWL-terrein is not always sufficient (participant 3), or acknowledged that certain groups may still need access to a car (participant 2). This was shared by one expert (K. Ruchtie & R. Sutmuller, personal communication, June 12, 2025), who expressed doubts at the feasibility and added benefits of going car-free versus low-car, due to e.g. transport of vulnerable groups who rely on cars.

This highlights a critical precondition for the successful implementation of car-free neighbourhoods: the availability of a dense, reliable, and accessible public transport network, which was also a condition identified by Nieuwenhuijsen et al. (2019).

These results highlight the crucial need for both qualitative public transport to serve as a feasible alternative to car usage, and ensuring destinations remain accessible by e.g. active mobility modes such as walking or cycling, as the latter means distances are kept small and therefore easier to traverse for vulnerable groups. From the GIS spatial analyses, it became apparent that Zaanstraat Emplacement is among the most suitable neighbourhoods in Haven-Stad for implementing a car-free design, ranking second in being least dependent on cars.

6.1 LIMITATIONS

While this study employed a mixed-methods approach to enhance the robustness of its findings, several limitations should be acknowledged. First, the thematic analysis that was employed is inherently interpretive and therefore susceptible to the bias of each individual researcher.

Factors such as prior knowledge, theoretical background, and assumptions may influence how data was interpreted and coded. Additionally, by relying on a deductive coding approach, which is guided by the concepts included in the research questions, a confirmation bias could potentially arise since predetermined codes are used, potentially limiting the discovery of new unexpected insights or patterns.

Second, another limitation stems from the GIS analysis and a lack of detailed spatial data. Although policy documents provide projections of the number of dwellings and workplaces programmed per neighbourhood, this information is relatively coarse as it is on a neighbourhood level, rather than, e.g. street or block level. As a result, it lacks the spatial resolution required for detailed accessibility modelling. To circumvent this issue, a methodological compromise was chosen, and randomly generated origin points that represent n number of dwellings or workplaces were used as inputs in the accessibility analysis as part of the car dependency index.

Third, while the multi-criteria analysis encompassed several mobility aspects, additional indicators, both transport-related and indirectly related to mobility, should be incorporated to increase the comprehensiveness of the analysis and better identify the optimal transport network.

Lastly, although the study discusses a broad range of interventions to support the development of a car-free neighbourhood, it may not have exhausted all possible solutions. Section 4.3 primarily offers recommendations based on existing literature and successful case studies. Still, it does not delve into the specific combination, implementation, or potential challenges and resistance that may arise during the implementation of these strategies in the context of Zaanstraat Emplacement. Therefore, despite proposing numerous measures worthy of consideration, further design is still needed to combine barriers, reasonable parking, and transit strategies in Zaanstraat, establishing the core elements of a car-free, pedestrian-friendly community.

6.2 FUTURE RESEARCH

The findings of this research contribute to a broader understanding of what factors influence a successful implementation of a car-free neighbourhood in the Dutch context by providing a detailed overview of those factors, as well as mobility scenario analyses on the Haven-Stad scale.

Additionally, this research analyses how accessibility, inclusivity, and well-connectedness can be balanced in a car-free neighbourhood design in redeveloping dense urban areas, broadening the knowledge about the relationship between these concepts.

Future research could aim to develop a more comprehensive understanding of mobility needs and perceptions by engaging a wider range of stakeholders, through e.g. purposive sampling. By employing this sampling method, a diversity in demographic and socio-economic backgrounds in the participant pool is ensured, thereby enriching the analysis. Additionally, the GIS analysis that aimed to assess the car dependency of neighbourhoods in Haven-Stad by analysing active mobility accessibility, density, and land-use function mix, relied on speculative data, such as projections of dwelling counts in a given neighbourhood. These numbers are subject to future changes, and therefore, the accuracy of these analyses could be significantly improved once more detailed and updated data becomes available.

7 CONCLUSION

This study aimed to explore to what extent Zaanstraat Emplacement can be developed as a car-free neighbourhood while remaining well-connected, accessible, and inclusive. In order to determine the feasibility and requirements for developing Zaanstraat Emplacement as a car-free neighbourhood, the conditions and barriers were explored though a SWOT analysis. Then the spatial mobility design needed to support a car-free environment was assessed, and finally relevant policy interventions and reference case to understand how car dependency can be effectively reduced were examined.

Achieving a successful car-free neighbourhood design in this context requires the integration of three key mobility planning principles, as identified in the theoretical framework and substantiated with our analysis findings: 1) well-connectedness, which refers to the integration of the neighbourhood into wider urban and regional mobility networks through fast, high-quality connections to other parts of the city; 2) accessibility, defined as the ability for all residents to reach key destinations within the neighbourhood through safe, direct, and reliable active mobility connections such as walking and cycling; 3) inclusivity, which ensures that the mobility network and public space are safe and usable for all people, regardless of age or socio-economic background. However, the SWOT analysis showed that barriers such as car dependency amongst residents and the rigidity of car-centric legislation must be addressed in order to successfully implement a car-free Zaanstraat Emplacement.

The potential of Zaanstraat Emplacement to be developed as a car-free neighbourhood is supported by the results from both the interviews that were conducted and the mobility impact analysis. Residents living in a car-free neighbourhood in Amsterdam express high satisfaction with their living environment, mentioning benefits such as increased safety, reduced noise, and more public greenery. This demonstrates that car-free neighbourhood design has already been successfully applied in the Amsterdam urban context with satisfactory results.

Additionally, the mobility impact analysis, which included the CDI, shows that Zaanstraat Emplacement ranks as the second most suitable neighbourhood within Haven-Stad for reducing car dependence.

Another critical factor in supporting a car-free lifestyle is high-quality and frequent public transport, which offers a reliable alternative to private car use. The MCA found that the most favourable scenario is an underground metro expansion through Houthavens, which is closer to the geometric centre of the Haven-Stad area and offers better connectivity to the walking and cycling networks, further strengthening the potential of Zaanstraat Emplacement for successful car-free development.

Finally, global reference cases highlighted that policy and governance solutions play a vital role in complementing physical interventions to stimulate a car-free lifestyle. For instance, communication strategies and campaigns can generate more awareness of the benefits and participation in car-free lifestyle.

To conclude, our findings show that Zaanstraat Emplacement has the potential to serve as a pilot site for implementing car-free neighbourhood design within Haven-Stad. Its spatial characteristics and planned transport infrastructure serve as favourable conditions to test car-free principles.

Based on the findings of this expert report, the following mobility interventions are recommended to enable a car-free development in Zaanstraat Emplacement. These will be elaborated upon along with a visualisation in the synthesis report.

- **Implementing 15-minute city principles through mixed-use development**
- **Providing shared mobility solutions for micromobility and motorised vehicles**
- **Organising behavioural change activities to encourage residents to enjoy a car-free community**
- **Use consent agreements for future residents to live in a car-free community.**
- **Closing the Amsterdam metro loop through Houthavens**
- **Building bridges over the remaining railyard for pedestrians and cyclists**

8 REFERENCES

Alessandretti, L., Natera Orozco, L. G., Saberi, M., Szell, M., & Battiston, F. (2023). Multimodal urban mobility and multilayer transport networks. Environment and Planning B: Urban Analytics and City Science, 50(8), 2038–2070. https://doi.org/10.1177/23998083221108190

Allam, Z., Khavarian-Garmsir, A. R., Lassaube, U., Chabaud, D., & Moreno, C. (2024). Mapping the Implementation Practices of the 15-Minute City. Smart Cities, 7(4), Article 4. https://doi.org/10.3390/smartcities7040083

Argent LLP. (2021). ULI Case Studies Kings Cross. ULI. https://database.dpa-etsam.com/wp-content/uploads/2021/03/kingscross_16pgs_v11.pdf

Barter, P. (2022, January 28). How the Netherlands handles parking. How the Netherlands Handles Parking. https://www.reinventingparking.org/2022/01/Netherlandsparking.html

Basu, R., & Ferreira, J. (2021). Planning car-lite neighborhoods: Does bikesharing reduce auto-dependence? Transportation Research Part D: Transport and Environment, 92, 102721. https://doi.org/10.1016/j.trd.2021.102721

Bertolini, L. (2017). Planning the Mobile Metropolis: Transport for People, Places and the Planet. Bloomsbury Publishing.

Brownrigg-Gleeson, M. L., Monzon, A., & Cortez, A. (2023). Reasons to Pedestrianise Urban Centres: Impact Analysis on Mobility Habits, Liveability and Economic Activities. Sustainability, 15(23), Article 23. https://doi.org/10.3390/su152316472

Cabanek, A., Zingoni de Baro, M. E., & Newman, P. (2020). Biophilic streets: A design framework for creating multiple urban benefits. Sustainable Earth, 3(1), 7. https://doi.org/10.1186/s42055-020-00027-0

Carvalho, J., De Sousa, J. P., & Macário, R. (2024). Towards a more inclusive mobility: Participatory mobility planning at a metropolitan scale. Transportation Research Procedia, 78, 222–229. https://doi.org/10.1016/j.trpro.2024.02.029

Caso, C. C. (2023). Exploring the Success Story of Culdesac, a Real Estate Development in Tempe, Arizona, a Case Study. ResearchGate. https://doi.org/10.13140/RG.2.2.25220.30084

Convention on the Rights of Persons with Disabilities, Pub. L. No. 45(1), A/RES/61/106 Human Rights Instruments 211 (2008).

Dehalwar, K., & Sharma, S. N. (2024). Exploring the Distinctions between Quantitative and Qualitative Research Methods. https://doi.org/10.5281/ZENODO.10553000

Ding, C., Wang, D., Liu, C., Zhang, Y., & Yang, J. (2017). Exploring the influence of built environment on travel mode choice considering the mediating effects of car ownership and travel distance. Transportation Research Part A: Policy and Practice, 100, 65–80. https://doi.org/10.1016/j.tra.2017.04.008

Dlabaja, C. (2023). Die Seestadt Aspern: Ein Stadtteil im Werden. Böhlau Wien.

Doheim, R. M., Farag, A., & Badawi, S. (2020). Success Measures for Transforming Into Car-Free Cities: Recommendations for Implementation (pp. 232–268). https://doi.org/10.4018/978-1-7998-3507-3.ch010

Drolsbach, S. (2025, June 6). Interview Simon Drolsbach Witteveen+Bos [Live interview].

Elfferding, S. (2025, June 27). Interview Referentin für Fußverkehr, Behörde für Verkehr und Mobilitätswende Hamburg [Live interview].

Equipaje, R. M. (2018). The Role of Public Space in Sustainable Urban Development. ResearchGate. https://doi.org/10.1007/978-3-319-57937-5_145

Foletta, N. (2011). Houten. ITDP Europe. https://www.houten.nl/fileadmin/user_upload/Over_gemeente_Houten/Pdf/FeitenCijfers2011/Europe_s_Vibrant_New_Low_Car_bon__Communities__Chapter_Houten.pdf

Frank, J., MacCleery, R., Nienaber, S., Hammerschmidt, S., Claflin, A., & Building Healthy Places Initiative (Eds.). (2015). Building healthy places toolkit: Strategies for enhancing health in the built environment. Urban Land Institute.

Gallo, M., & Marinelli, M. (2020). Sustainable Mobility: A Review of Possible Actions and Policies. Sustainability, 12(18), Article 18. https://doi.org/10.3390/su12187499

Gehl, J. (2013). Cities for People. Island Press.

Gemeente Amsterdam. (n.d.). Frans Halsbuurt: Opnieuw inrichten [Webpagina]. Amsterdam.nl; Gemeente Amsterdam. Retrieved July 2, 2025, from https://www.amsterdam.nl/projecten/franshalsbuurt/

Gemeente Amsterdam. (2017). Ontwikkelstrategie Haven-Stad. https://openresearch.amsterdam/image/2020/7/1/ontwikkelstrategie_haven_stad_30_mei_2017.pdf

Gemeente Amsterdam. (2021a). Uitvoeringsprogramma Haven-Stad 2021-2025. https://openresearch.amsterdam/nl/page/79580/uitvoeringsprogramma-haven-stad-2021-2025

Gemeente Amsterdam. (2021b, November 8). Integraal Raamwerk Haven-Stad. https://openresearch.amsterdam/image/2022/1/10/integraal_raamwerk_haven_stad_november_2021.pdf

Gemeente Utrecht. (2021). Omgevingsvisie Merwedekanaalzone, Deel 2: Uitwerking van de Ruimtelijke Agenda.

Gerike, R., Koszowski, C., Schröter, B., Buehler, R., Schepers, P., Weber, J., Wittwer, R., & Jones, P. (2021). Built Environment Determinants of Pedestrian Activities and Their Consideration in Urban Street Design. Sustainability, 13(16), Article 16. https://doi.org/10.3390/su13169362

Geurs, K. T., & van Wee, B. (2004). Accessibility evaluation of land-use and transport strategies: Review and research directions. Journal of Transport Geography, 12(2), 127–140. https://doi.org/10.1016/j.jtrangeo.2003.10.005

Glazener, A., Wylie, J., van Waas, W., & Khreis, H. (2022). The Impacts of Car-Free Days and Events on the Environment and Human Health. Current Environmental Health Reports, 9(2), 165–182. https://doi.org/10.1007/s40572-022-00342-y

Greeve, W. (2025, June 6). Interview APPM [Live interview].

Gupta, A., Yadav, M., & Nayak, B. K. (2025). A Systematic Literature Review on Inclusive Public Open Spaces: Accessibility Standards and Universal Design Principles. Urban Science, 9(6), Article 6. https://doi.org/10.3390/urbansci9060181

GWL Terrein. (2024). Ontstaan van de wijk. GWL Terrein. https://gwl-terrein.nl/de-wijk/ontstaan-van-de-wijk/

Harries, A. (2025, June 17). Interview Related Argent Kings Cross expert [Live interview].

Hensher, D. A. (2008). Climate change, enhanced greenhouse gas emissions and passenger transport – What can we do to make a difference? Transportation Research Part D: Transport and Environment, 13(2), 95–111. https://doi.org/10.1016/j.trd.2007.12.003

Heyken Soares, P. (2021). Zone-based public transport route optimisation in an urban network. Public Transport, 13(1), 197–231. https://doi.org/10.1007/s12469-020-00242-0

Horiike, T., Yoh, K., Doi, K., & Chou, C.-C. (2023). Assessing the Hierarchical Diversity of Public Transportation Considering Connectivity and Its Implication on Regional Sustainability. Sustainability, 15(23), Article 23. https://doi.org/10.3390/su152316494

Iannillo, A., & Fasolino, I. (2021). Land-Use Mix and Urban Sustainability: Benefits and Indicators Analysis. Sustainability, 13(23), Article 23. https://doi.org/10.3390/su132313460

Jorritsma, P. (2023). Autoluw beleid gemeenten. Kennisinstituut voor Mobiliteitsbeleid.

Khreis, H., & Nieuwenhuijsen, M. (2016). Car free cities: Pathway to healthy urban living. Environment International, 94, 251–262. https://doi.org/10.1016/j.envint.2016.05.032

Kim, J., & Brownstone, D. (2013). The impact of residential density on vehicle usage and fuel consumption: Evidence from national samples. Energy Economics, 40, 196–206. https://doi.org/10.1016/j.eneco.2013.06.012

Krukowicz, T., Firląg, K., Sobota, A., Kołodziej, T., & Novačko, L. (2021). The relationship between bicycle traffic and the development of bicycle infrastructure on the example of Warsaw. Archives of Transport, 60(4), Article 4. https://doi.org/10.5604/01.3001.0015.6930

Kushner, J. A. (2005). Car-Free Housing Developments: Toward Sustainable Smart Growth and Urban Regeneration Through Car-Free Zoning, Car-Free Redevelopment, Pedestrian Improvement Districts, and New Urbanism. ResearchGate. https://doi.org/10.5070/L5231019795

Lonardi, A., Szell, M., & Bacco, C. D. (2025). Cohesive urban bicycle infrastructure design through optimal transport routing in multilayer networks. Journal of The Royal Society Interface, 22(223). https://doi.org/10.1098/rsif.2024.0532

Luoto, L., & Lonkila, A. (2018). The Use of SWOT Analysis for Future Scenarios: A Case Study of Privacy and Emerging Technologies. In R. Peperhove, K. Steinmüller, & H.-L. Dienel (Eds.), Envisioning Uncertain Futures: Scenarios as a Tool in Security, Privacy and Mobility Research (pp. 105–131). Springer Fachmedien. https://doi.org/10.1007/978-3-658-25074-4_7

Mackett, R. L., & Thoreau, R. (2015). Transport, social exclusion and health. Journal of Transport & Health, 2(4), 610–617. https://doi.org/10.1016/j.jth.2015.07.006

Mackey, A., & Bryfonski, L. (2018). Mixed Methodology. In A. Phakiti, P. De Costa, L. Plonsky, & S. Starfield (Eds.), The Palgrave Handbook of Applied Linguistics Research Methodology (pp. 103–121). Palgrave Macmillan UK. https://doi.org/10.1057/978-1-137-59900-1_5

Magdin, K., Mavrin, V., & Boyko, A. (2019). Correlation between Noise and Air Pollution from Car Sources. 2019 12th International Conference on Developments in eSystems Engineering (DeSE), 812–816. https://doi.org/10.1109/DeSE.2019.00151

Marzolla, F., Bruno, M., Melo, H. P. M., & Loreto, V. (2024). Compact 15-minute cities are greener (No. arXiv:2409.01817). arXiv. https://doi.org/10.48550/arXiv.2409.01817

McAskie, L. (2021, January 18). Lessons From Amsterdam's Agenda—Taking "Car-Lite" to the Next Level. CityChangers.Org – Home Base for Urban Shapers. https://citychangers.org/lessons-from-amsterdams-agenda/

Melia, S. (2014). Carfree and Low-Car Development. In Parking Issues and Policies (world; Vol. 5, pp. 213–233). Emerald Group Publishing Limited. https://doi.org/10.1108/S2044-994120140000005012

Mishra, P. (2020). Multi Modal Transit Hub as a Solution for Growing Urban Traffic Congestion in Mega Cities. https://www.ijsr.net/archive/v9i6/SR20627015819.pdf

Ng, E. (2010). Designing High-Density Cities: For Social and Environmental Sustainability. Earthscan.

Nielsen, G. (2006, April 1). Integration or segregation—Recommendations of the principles of urban road network design for the sustainable city. SWOV. https://swov.nl/nl/publicatie/integration-or-segregation-recommendations-principles-urban-road-network-design

Nielsen, Olafsson, A. S., Carstensen, T. A., & Skov-Petersen, H. (2013). Environmental correlates of cycling: Evaluating urban form and location effects based on Danish micro-data. Transportation Research Part D: Transport and Environment, 22, 40–44. https://doi.org/10.1016/j.trd.2013.02.017

Nieuwenhuijsen, M., Bastiaanssen, J., Sersli, S., Waygood, E. O. D., & Khreis, H. (2019). Implementing Car-Free Cities: Rationale, Requirements, Barriers and Facilitators. In Integrating Human Health into Urban and Transport Planning (pp. 199–219). Springer, Cham. https://doi.org/10.1007/978-3-319-74983-9_11

Nunen, J. A. E. E. van, Huijbregts, P., & Rietveld, P. (2011). Transitions Towards Sustainable Mobility: New Solutions and Approaches for Sustainable Transport Systems. Springer Science & Business Media.

OpenStreetMap. (n.d.). OpenStreetMap. OpenStreetMap. Retrieved June 25, 2025, from https://www.openstreetmap.org/

Parolek, D. (2025, January 2). Mixed-Use Neighborhood Designed Car-Free. Greater Phoenix In Business Magazine. https://inbusinessphx.com/commercial-real-estate/mixed-use-neighborhood-designed-car-free

Prados, M.-J., Iglesias-Pascual, R., & Barral, Á. (2022). Energy transition and community participation in Portugal, Greece and Israel: Regional differences from a multi-level perspective. Energy Research & Social Science, 87, 102467. https://doi.org/10.1016/j.erss.2021.102467

Quiroz Villanueva, M. J., Guglielmi, F., De Fabiis, F., & Coppola, P. (2025). Social and Economic Impacts of Transportation Multi-Modal and Multi-Service Hub Development. Sustainability, 17(4), Article 4. https://doi.org/10.3390/su17041767

Reggiani, G., Van Oijen, T., Hamedmoghadam, H., Daamen, W., Vu, H. L., & Hoogendoorn, S. (2022). Understanding bikeability: A methodology to assess urban networks. Transportation, 49(3), 897–925. https://doi.org/10.1007/s11116-021-10198-0

Ruchtie, K., & Sutmuller, R. (2025, June 12). Interview OVSA [Online interview].

Rydningen, U. (2017). OSLO 2019: A CAR-FREE CITY CENTRE. ResearchGate. SUSTAINABLE DEVELOPMENT AND PLANNING 2017. https://doi.org/10.2495/SDP170011

Rye, T., Burns, J., Whitefield, A., Oxley-Glenister, K., & Pinkard, J. (2023). Reducing car use through parking policies: An evidence review. http://dx.doi.org/10.7488/era/3776

Saeidizand, P., Fransen, K., & Boussauw, K. (2022). Revisiting car dependency: A worldwide analysis of car travel in global metropolitan areas. Cities, 120, 103467. https://doi.org/10.1016/j.cities.2021.103467

Schimek, P. (1996). Household Motor Vehicle Ownership and Use: How Much Does Residential Density Matter? Transportation Research Record, 1552(1), 120–125. https://doi.org/10.1177/0361198196155200117

Stavroulaki, I., & Pont, M. B. (2020). A SYSTEMATIC REVIEW OF MULTIFUNCTIONAL STREETS_Final research report. https://doi.org/10.13140/RG.2.2.21921.63840

Steinacker, C., Storch, D.-M., Timme, M., & Schröder, M. (2022). Demand-driven design of bicycle infrastructure networks for improved urban bikeability. Nature Computational Science, 2(10), 655–664. https://doi.org/10.1038/s43588-022-00318-w

Steuteville, R. (2024, September 6). Lessons from ‘car-free’ Culdesac [Text]. CNU. https://www.cnu.org/publicsquare/2024/09/06/lessons-%E2%80%98car-free%E2%80%99-culdesac

Sun, B., Ermagun, A., & Dan, B. (2017). Built environmental impacts on commuting mode choice and distance: Evidence from Shanghai. Transportation Research Part D: Transport and Environment, 52, 441–453. https://doi.org/10.1016/j.trd.2016.06.001

Takahashi, K., & Maeno, T. (2011). The Causal SWOT Analysis Using Systems Thinking As A Tool for Situational Analysis Workshop. Proceedings of the 55th Annual Meeting of the ISSS - 2011, Hull, UK. https://journals.isss.org/index.php/proceedings55th/article/view/1663

Tang, J., Heinimann, H., Han, K., Luo, H., & Zhong, B. (2019). Evaluating resilience in urban transportation systems for sustainability: A systems-based Bayesian network model (No. arXiv:1908.09774). arXiv. https://doi.org/10.48550/arXiv.1908.09774

Ton, D., Cats, O., Duives, D., & Hoogendoorn, S. (2017). How do people cycle in Amsterdam? Estimating cyclists’ route choice determinants using GPS data from an urban area. The 96th Annual Meeting of the Transportation Research Board, 1–12.

Tufail, A., Moeinaddini, M., Almoshaogeh, M., Jamal, A., Nawaz, I., & Alharbi, F. (2021). A New Pedestrian Crossing Level of Service (PCLOS) Method for Promoting Safe Pedestrian Crossing in Urban Areas. Int. J. Environ. Res. Public Health. https://doi.org/10.3390/ijerph18168813

van der Bijl, R., & Zeeuw, F. de. (2009). Integrating public transport into urban area development. World Transit Research. https://www.worldtransitresearch.info/research/2776

van Dort, A. (2022). Haven-Stad: Duurzame stad van overmorgen. openresearch.amsterdam. https://openresearch.amsterdam/nl/page/84983/haven-stad-duurzame-stad-van-overmorgen
Verkade, T., & Brömmelstroet, M. te. (2024). Movement: How to Take Back Our Streets and Transform Our Lives. Island Press.

Walvius, M. (2025, June 11). Interview MADE Zaanstraat Emplacement Autovrij [Online interview].

Wendling, L., Rinta-Hiiro, V., Jermakka, J., Fatima, Z., & zu-Castell Rüdenhausen, M. (2020). Performance and Impact Monitoring of Nature-Based Solutions. UNaLab. https://unalab.eu/system/files/2020-02/d31-nbs-performance-and-impact-monitoring-report2020-02-17.pdf

WSPP. (2025). The Western Sydney Neighbourhoods Project [Discussion Paper]. https://theparks.nsw.gov.au/wp-content/uploads/2025/03/The-Western-Sydney-Neighbourhoods-Project-Discussion-Paper.pdf